RAG Pipeline Design and Vector Databases

Scaler School of Technology | ML System Design and LLMOps | Week 4 — Session 1/2 | Pre-Lecture Prep

What Is This Lecture About?

Large language models are powerful, but they have a fixed knowledge cutoff, they cannot see your private data, and they hallucinate facts they were never trained on. Retrieval-Augmented Generation (RAG) is the industry-standard architecture for solving these problems. Instead of fine-tuning a model on every new document, RAG retrieves relevant chunks at inference time and injects them into the prompt — so the model reasons over real, current, sourced information.

This session covers the full RAG pipeline, from document ingestion and chunking to vector databases, hybrid search, and evaluation. This is production LLMOps knowledge from 2023–2024, directly applicable to your projects.

Why RAG Exists and When to Use It

LLM limitations: training cutoff, hallucination, no citations. When to choose RAG over fine-tuning over prompt engineering — and why fine-tuning is often the *wrong* tool for adding factual knowledge.

Chunking and Embedding

How to split documents into chunks (fixed, sentence, recursive, semantic), why chunk size is a critical hyperparameter, and why the embedding model version at index time must match query time.

Vector Databases and Hybrid Search

What a vector DB does, HNSW vs. IVF approximate nearest neighbour algorithms, and which database to pick for your use case. Why hybrid search (dense + sparse + RRF) beats pure embedding search in production.

Evaluation and Failure Modes

RAGAS metrics: faithfulness, answer relevancy, context precision, context recall. The five most common RAG failure modes and the guardrails that prevent them.

What to Expect in the Session

- Opening: why does your LLM not just "know everything"?
- Full RAG architecture diagram walkthrough (indexing + query pipeline)
- Two short quizzes: chunking strategy design and failure mode diagnosis
- Vector DB comparison table and decision framework
- Hybrid search (BM25 + dense) and RRF explained with an example
- Live demo: build a mini RAG pipeline over a small document set
- RAGAS evaluation walkthrough: measuring what actually matters

How to Prepare

- Think about a use case where a chatbot gave you a wrong or outdated answer. What would it have needed to answer correctly?
- Recall what a cosine similarity score means — we will use it extensively
- Optional: skim the Weaviate blog post "What is RAG?" (free, 10 minutes)
- **Install before the session:**

```
pip install langchain chromadb
pip install sentence-transformers
```

`pip install ragas`

- Ensure you have a Google Colab runtime with GPU (needed for embedding model demo)
- No OpenAI API key required — we will use open-source embedding models

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Duration: ∼2 hours **Prerequisites:** Weeks 1–3 of this course; familiarity with Python and basic NLP **Instructor:** Priyansh Saxena